IBM APPLIED DATA SCIENCE CAPSTONE

Winning the Space Race with Data Science

Predicting SpaceX Falcon 9 First-Stage Landing Success

xwarcriminalx | July 2026

Project repository: https://github.com/xwarcriminalx/IDM_Data_Coursera

Executive Summary

This is the final **Data Science Capstone Project Report** submitted as a PDF presentation. It analyses SpaceX Falcon 9 launch data to predict whether the first stage will land successfully. The complete project code - **all completed Jupyter notebooks and the Plotly Dash Python file** - is published in the project GitHub repository:

https://github.com/xwarcriminalx/IDM_Data_Coursera

Methodologies used

- **Data collection** from the SpaceX REST API and by **web scraping** Wikipedia with BeautifulSoup.
- **Data wrangling**: cleaning, one-hot encoding and building the binary **Class** landing label.
- **EDA** with Pandas / Matplotlib / Seaborn and with **SQL**; interactive **Folium** maps and a **Plotly Dash** dashboard.
- **Predictive modelling**: Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV.

Key results

- Overall first-stage landing success rate is **67%** and has **improved strongly since 2013**.
- Success depends on **launch site, orbit type and payload mass**; KSC LC-39A and low payloads land most reliably.
- All four tuned classifiers reach **83.3% test accuracy**; the **Decision Tree** is best (validation 0.86).

Introduction - Background & Problem

Background

SpaceX advertises Falcon 9 launches at **$62 million**, far cheaper than competitors (up to $165 million), largely because it **reuses the first stage**. If we can predict whether the first stage will land successfully, we can estimate the true cost of a launch.

This is valuable for a competing company (SpaceY) bidding against SpaceX.

Problems to answer

- What factors determine whether the first stage lands successfully?
- How do launch site, orbit, payload mass and previous flights affect landing success?
- What conditions give the best chance of a successful landing?
- Can we build a model that **predicts** first-stage landing outcome?

Methodology - Workflow Overview

- **Data collection** - SpaceX REST API (requests -> pd.json_normalize) and web scraping Wikipedia (BeautifulSoup).
- **Data wrangling** - filter to Falcon 9, handle missing values, engineer the **Class** landing label.
- **EDA** - visualization (scatter / bar / line) and SQL queries.
- **Interactive visual analytics** - Folium maps and a Plotly Dash dashboard.
- **Predictive analysis** - build, tune (GridSearchCV) and evaluate classification models; compare with a confusion matrix.

Data Collection - SpaceX API

Process (key phrases / flowchart)

- **requests.get()** the SpaceX API /launches/past endpoint.
- Response **.json()** -> **pd.json_normalize()** to a dataframe.
- Use rocket / payload / launchpad / core IDs to call helper endpoints and enrich columns (BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, GridFins, Reused, Legs, LandingPad, Block, Serial, Longitude, Latitude).
- Filter to **Falcon 9**; replace missing PayloadMass with the mean.
- Export the cleaned dataframe to CSV.

Outcome

A clean dataframe of **90 Falcon 9 launches** with the features used throughout the project.

GitHub notebook: https://github.com/xwarcriminalx/IDM_Data_Coursera/blob/main/jupyter-labs-spacex-data-collection-api.ipynb

Data Collection - Web Scraping

Process (key phrases / flowchart)

- **requests.get()** the Wikipedia 'List of Falcon 9 and Falcon Heavy launches' page.
- Create a **BeautifulSoup** object from the HTML response.
- Find all <table> elements; locate the launch-records table.
- Extract column names from the header cells.
- Iterate rows and parse each cell into a dictionary of launch records.
- Build a Pandas dataframe and export to CSV.

Outcome

A dataframe of launch records (Flight No., Date, Booster, Launch site, Payload, Orbit, Customer, Outcome) parsed from the HTML tables.

GitHub notebook:
https://github.com/xwarcriminalx/IDM_Data_Coursera/blob/main/jupyter-labs-webscraping.ipynb

Data Wrangling Methodology

Cleaning & processing steps

- Computed launches per site, per orbit and per landing outcome with value_counts().
- Analyzed the **Outcome** column (True/False ASDS, RTLS, Ocean, None).
- Created a binary **training label Class**: 1 = successful landing, 0 = unsuccessful.
- Verified the resulting **success rate = 0.667**.

Outcome

A tidy dataset where every launch has a clear numeric **Class** target ready for EDA and modelling.

GitHub notebook: https://github.com/xwarcriminalx/IDM_Data_Coursera/blob/main/labs-jupyter-spacex-Data-wrangling.ipynb

EDA with Data Visualization - Methodology

Charts used and why

- **Scatter plots** (Flight No. vs Site, Payload vs Site, Flight No. vs Orbit, Payload vs Orbit) reveal how relationships between variables affect success.
- **Bar chart** (success rate by orbit) compares a categorical variable against the success rate.
- **Line chart** (success rate by year) shows the trend of success over time.

Outcome

Identified that **launch site, orbit and payload mass** influence landing success, and that success **trends upward over the years**.

GitHub notebook: https://github.com/xwarcriminalx/IDM_Data_Coursera/blob/main/jupyter-labs-eda-dataviz.ipynb

EDA with SQL - Methodology

SQL queries performed

- Listed unique launch sites and inspected CCA records.
- Aggregated payload mass (total for NASA CRS, average for F9 v1.1).
- Found the first successful ground-pad landing date.
- Filtered boosters by drone-ship success and payload range; found max-payload boosters.
- Counted mission outcomes and **ranked landing outcomes** in a date window.

Outcome

Loaded data into SQLite and answered business questions with DISTINCT, SUM/AVG, LIKE, BETWEEN, GROUP BY and subqueries.

GitHub notebook: https://github.com/xwarcriminalx/IDM_Data_Coursera/blob/main/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Interactive Visual Analytics - Methodology

Folium map

- Marked all launch sites; added colour-coded success/failure markers with MarkerCluster.
- Drew **proximity lines** and distances from a launch site to the nearest coastline.

Plotly Dash dashboard

- Dropdown to select launch site; **pie chart** of success and **scatter** of payload vs outcome with a payload range slider.

Outcome

Interactive tools that let stakeholders explore launch geography and the payload / booster factors behind success.

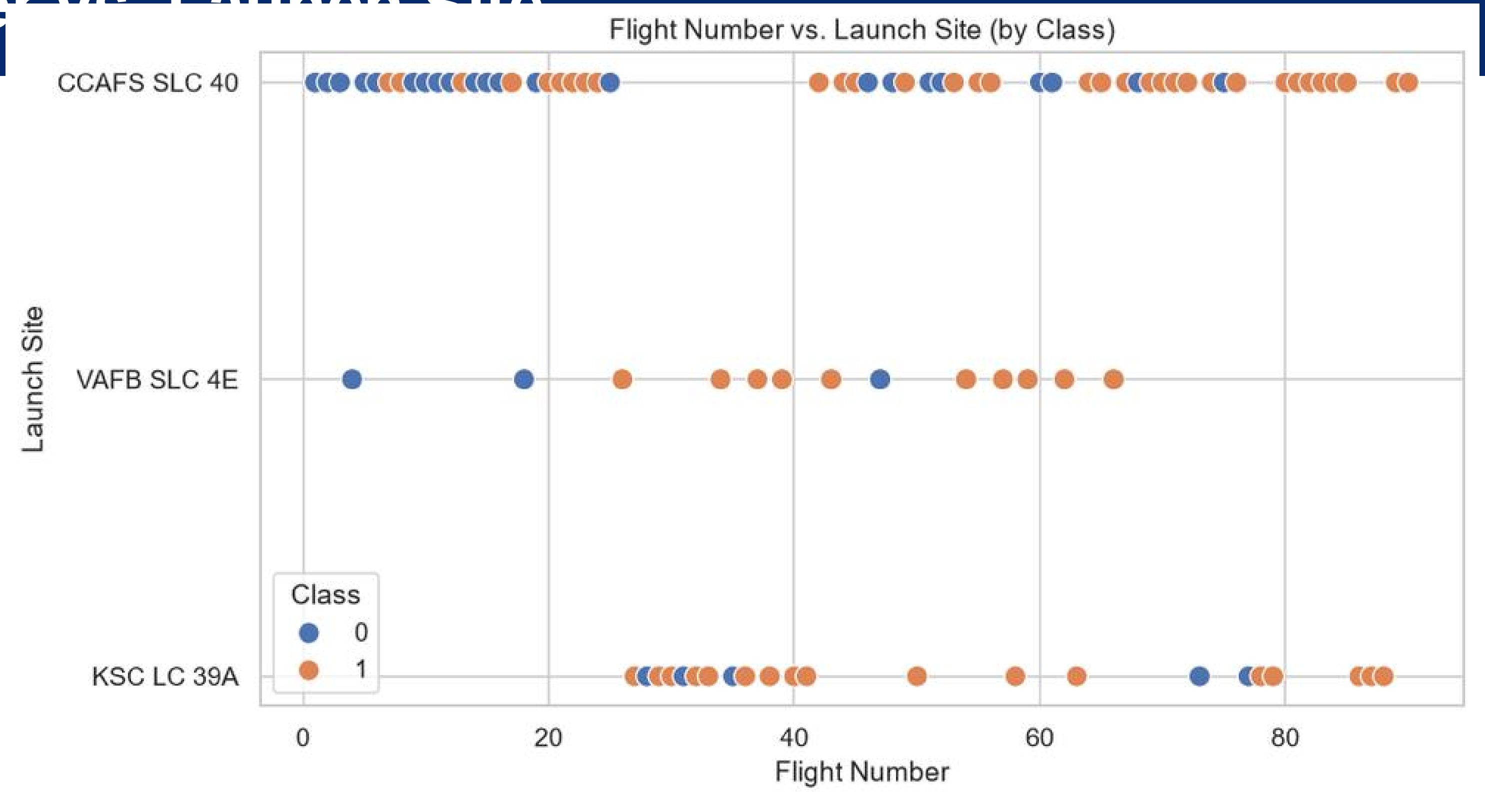
Folium: https://github.com/xwarcriminalx/IDM_Data_Coursera/blob/main/lab_jupyter_launch_site_location.ipynb

Dash app: https://github.com/xwarcriminalx/IDM_Data_Coursera/blob/main/spacex_dash_app.py

Predictive Analysis - Methodology

- **Prepare** - standardize features (StandardScaler); split train/test with test_size=0.2, random_state=2 -> **18 test records**.
- **Build & tune** - Logistic Regression, SVM, Decision Tree and KNN, each tuned with GridSearchCV(cv=10).
- **Evaluate** - compare validation (CV) and test accuracy; inspect the **confusion matrix** of the best model.
- **Select** - choose the model with the best performance.

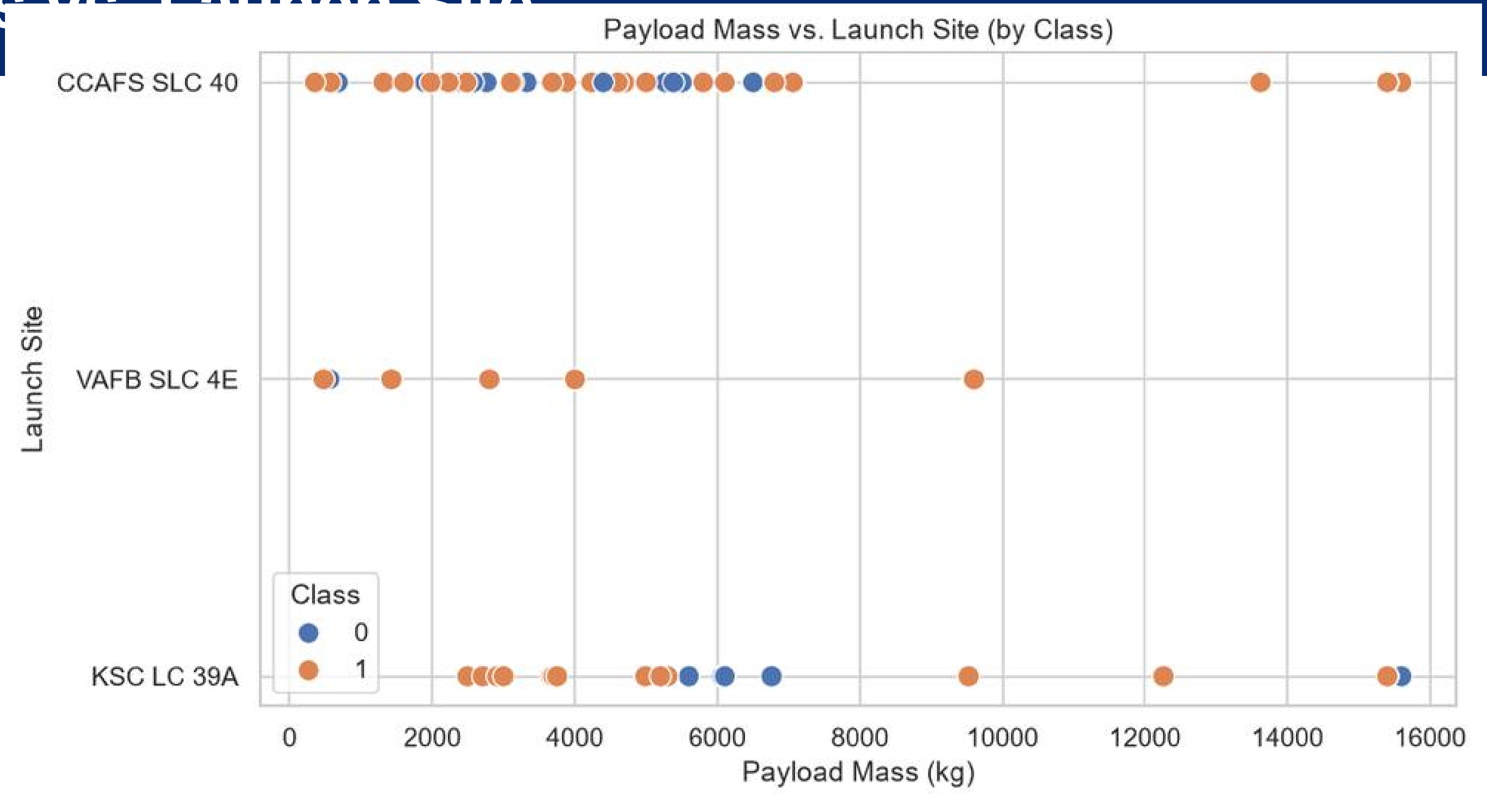
Flight Number vs. Launch Site



Insight: CCAFS SLC-40 hosts most early launches; successes (orange) become more frequent at higher flight numbers.

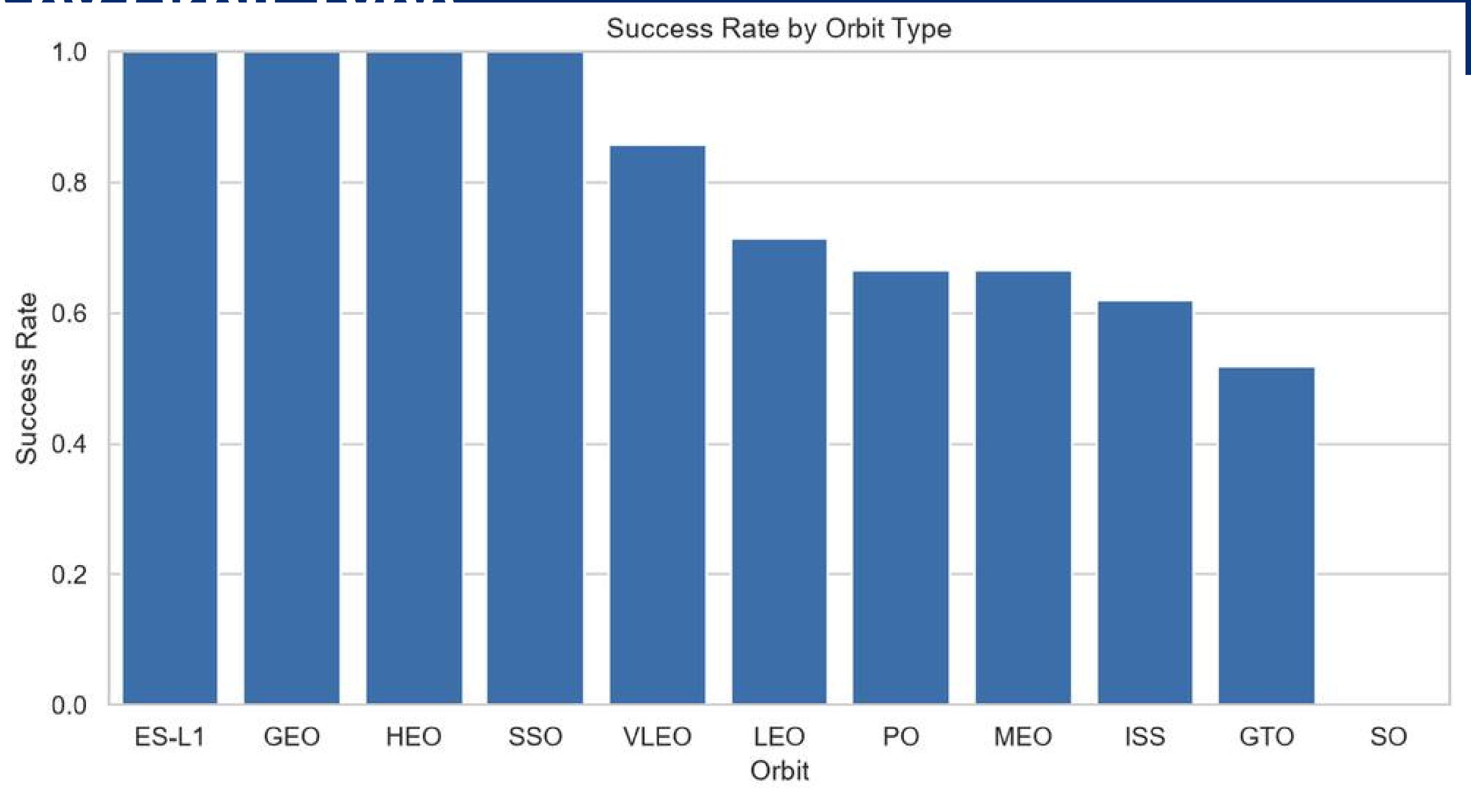
RESULTS - EDA VISUALIZATION

Payload Mass vs. Launch Site



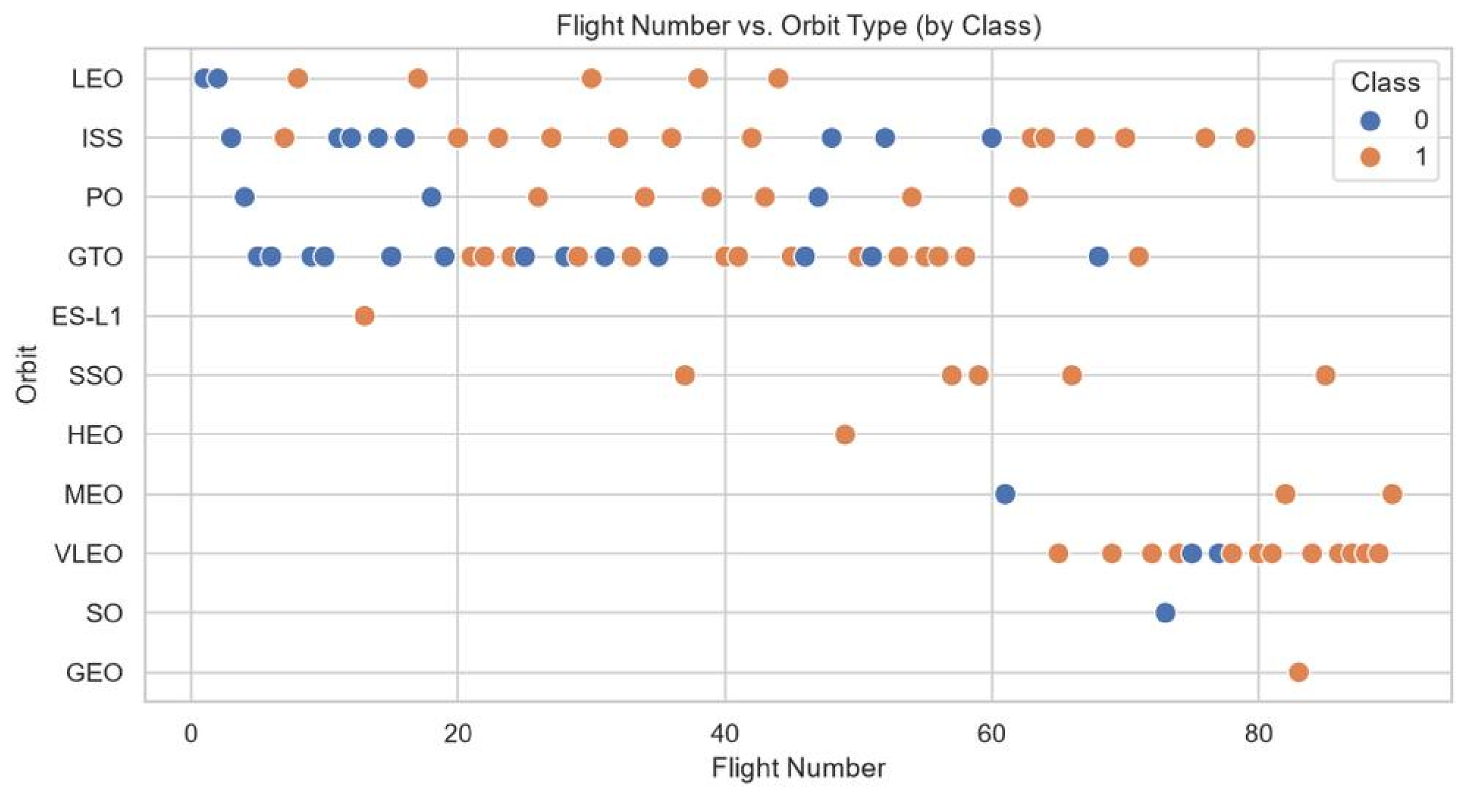
Insight: VAFB SLC-4E has no launches above ~10,000 kg; heavier payloads still land successfully at CCAFS and KSC.

Success Rate by Orbit Type



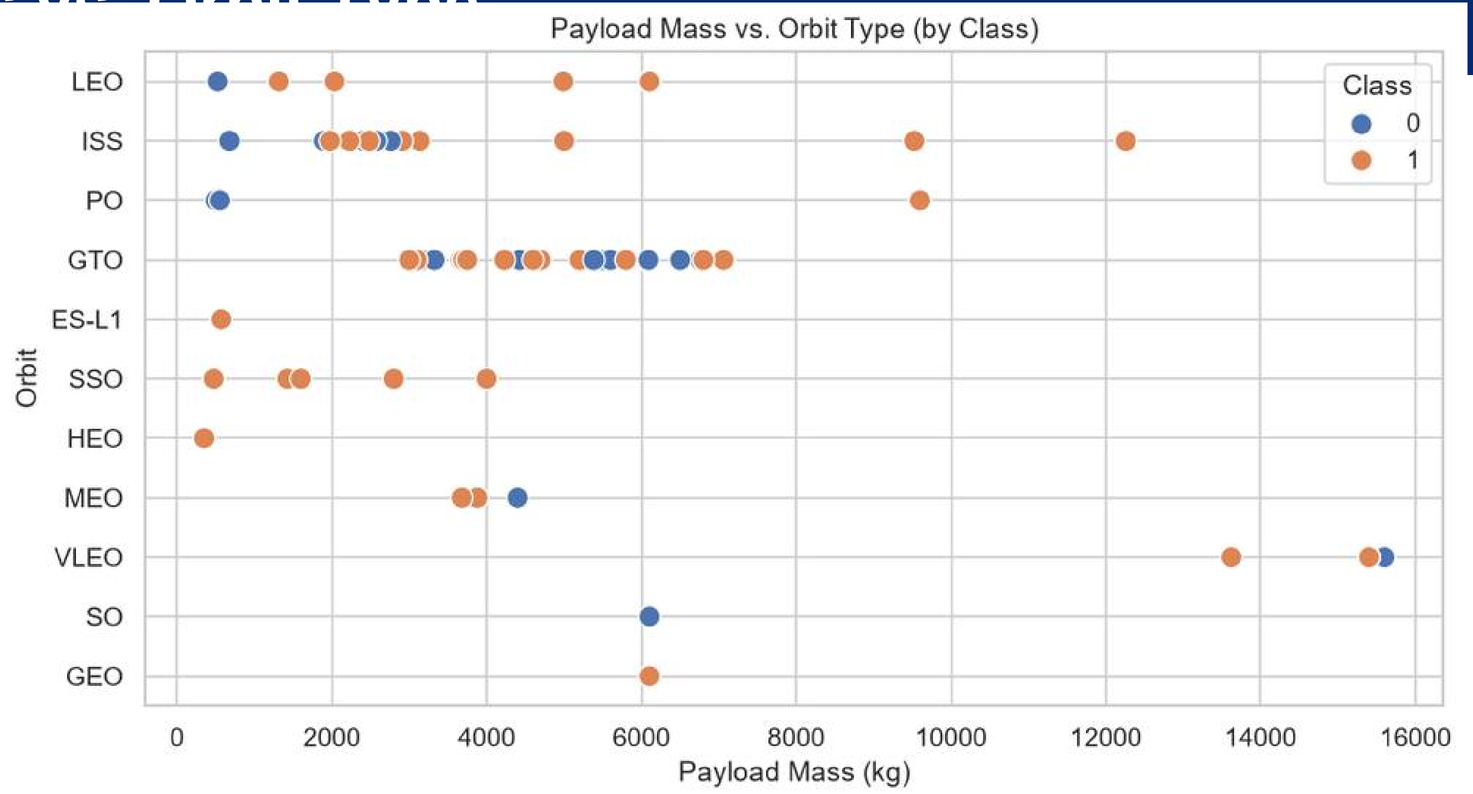
Insight: ES-L1, GEO, HEO and SSO show the highest success rates (~1.0); GTO is among the lowest.

Flight Number vs. Orbit Type



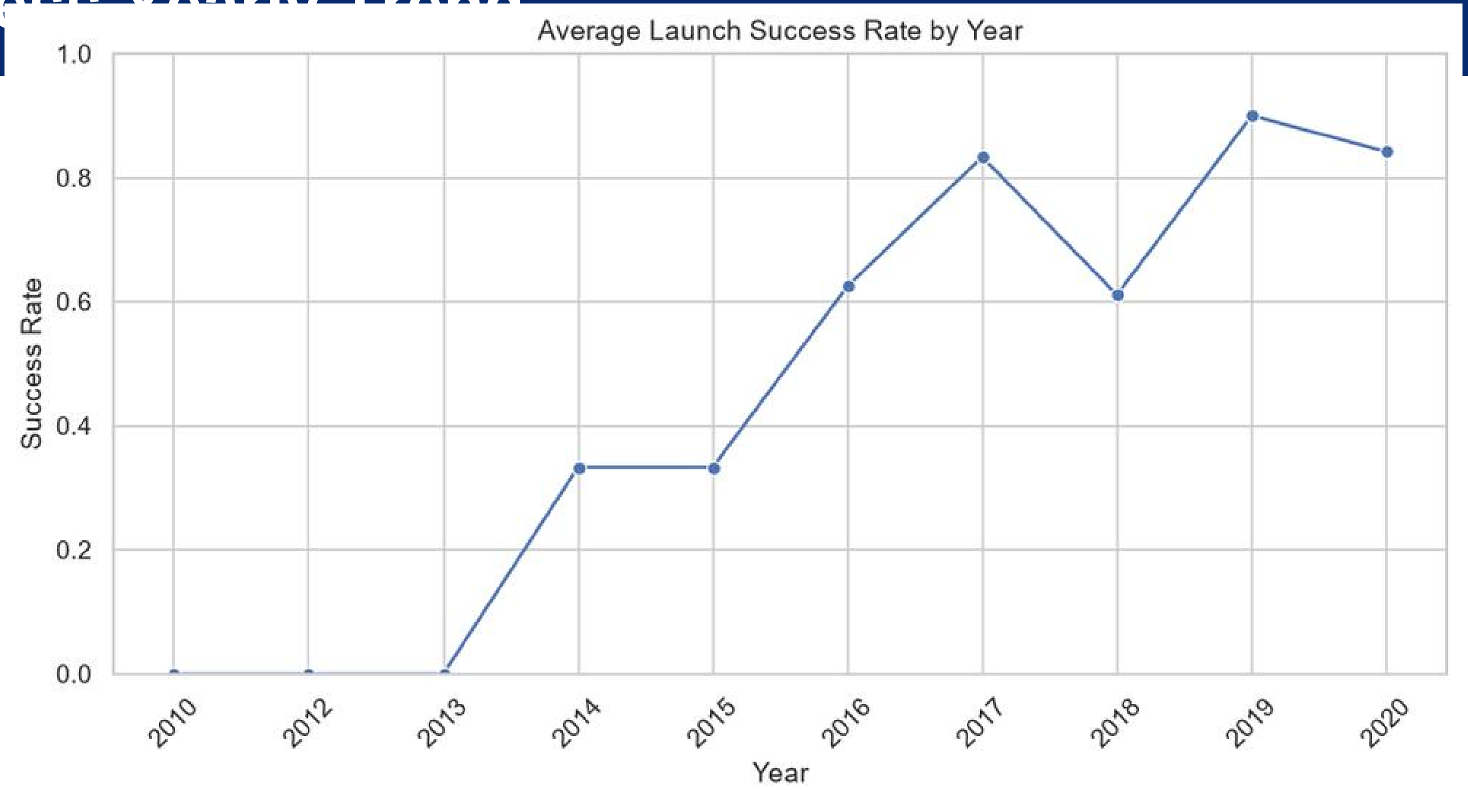
Insight: For LEO, success relates to the number of flights; in GTO there is no clear relationship with flight number.

Payload Mass vs. Orbit Type



Insight: Heavy payloads land more successfully for PO, LEO and ISS; for GTO the effect of payload is mixed.

Launch Success Yearly Trend



Insight: Success rate was near zero before 2013 and climbs steadily afterwards, reaching high levels by 2019-2020.

SQL Query Results

1. Unique launch sites

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

3. Total payload mass by NASA (CRS)

Total_Payload_NASA_CRS
45596

SQL Query Results

4. Average payload mass by booster F9 v1.1

Avg_Payload_F9_v1_1
2928.4

5. First successful ground pad landing date

First_Successful_Ground_Pad
2015-12-22

SQL Query Results

6. Boosters: success on drone ship & payload 4000-6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

7. Count of successful vs failed mission outcomes

Mission_Outcome	Total
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

SQL Query Results

9. 2015 failed drone-ship landings (month names)

Month	Booster_Version	Launch_Site
January	F9 v1.1 B1012	CCAFS LC-40
April	F9 v1.1 B1015	CCAFS LC-40

10. Rank landing outcomes 2010-06-04 to 2017-03-20

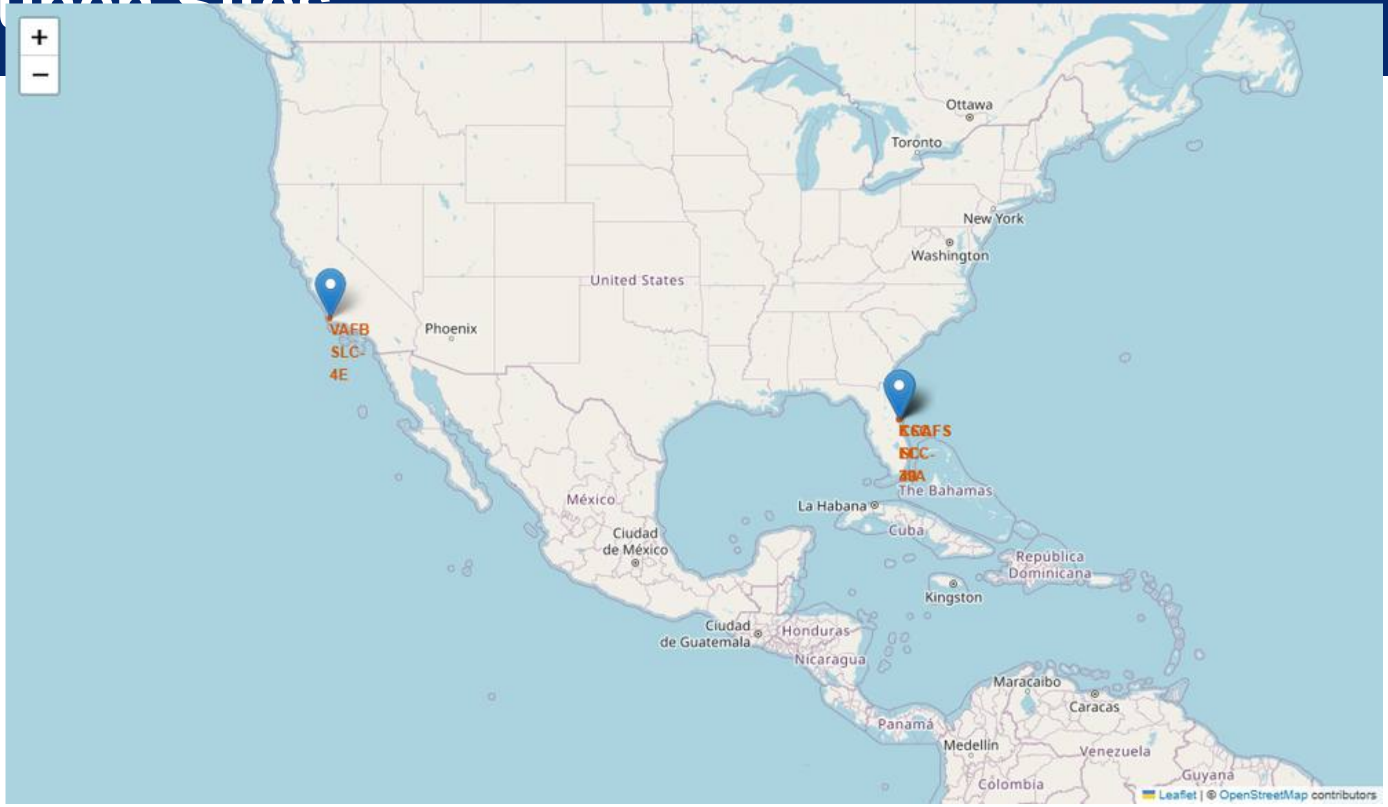
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

SQL Query Results - CCA Launch Records

2. 5 records where launch site begins with "CCA"

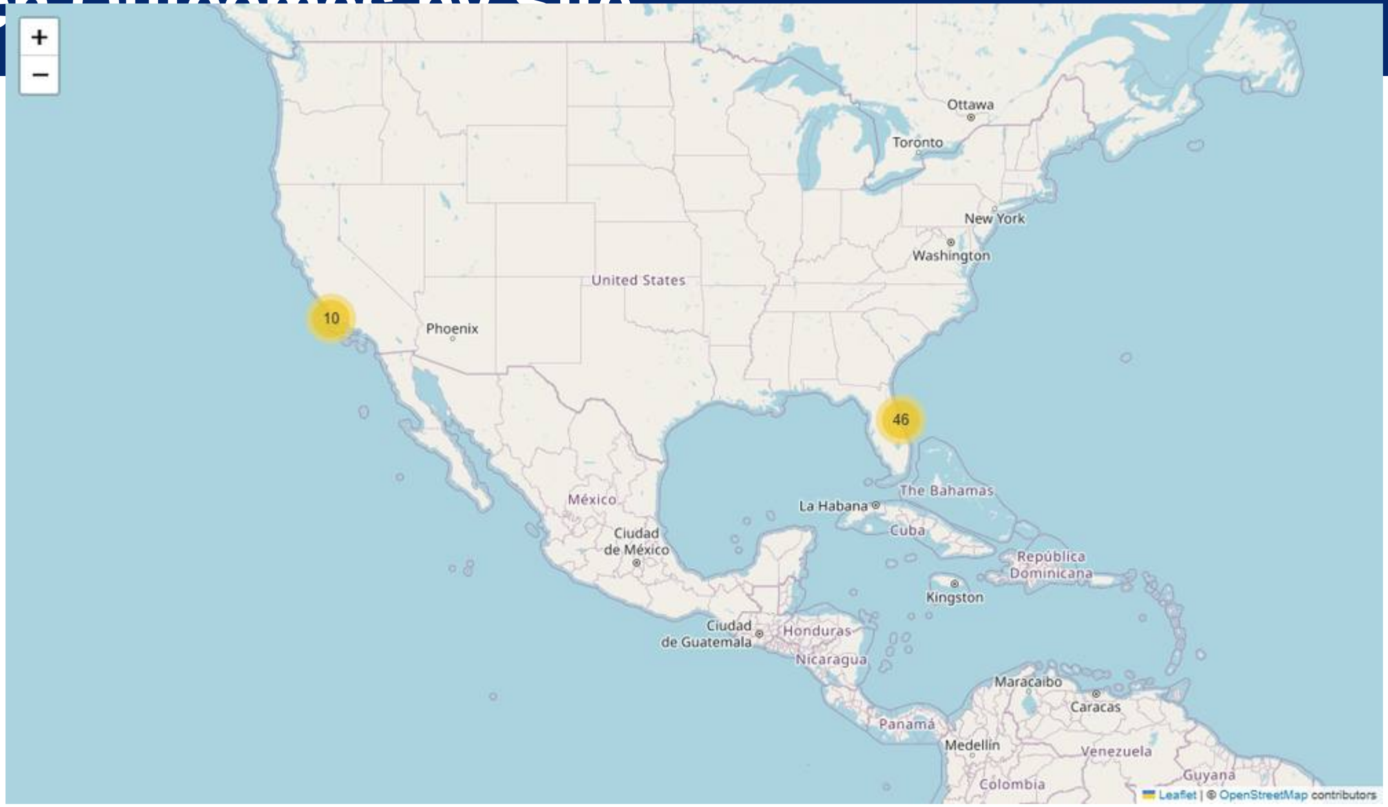
Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Folium - All Launch Sites



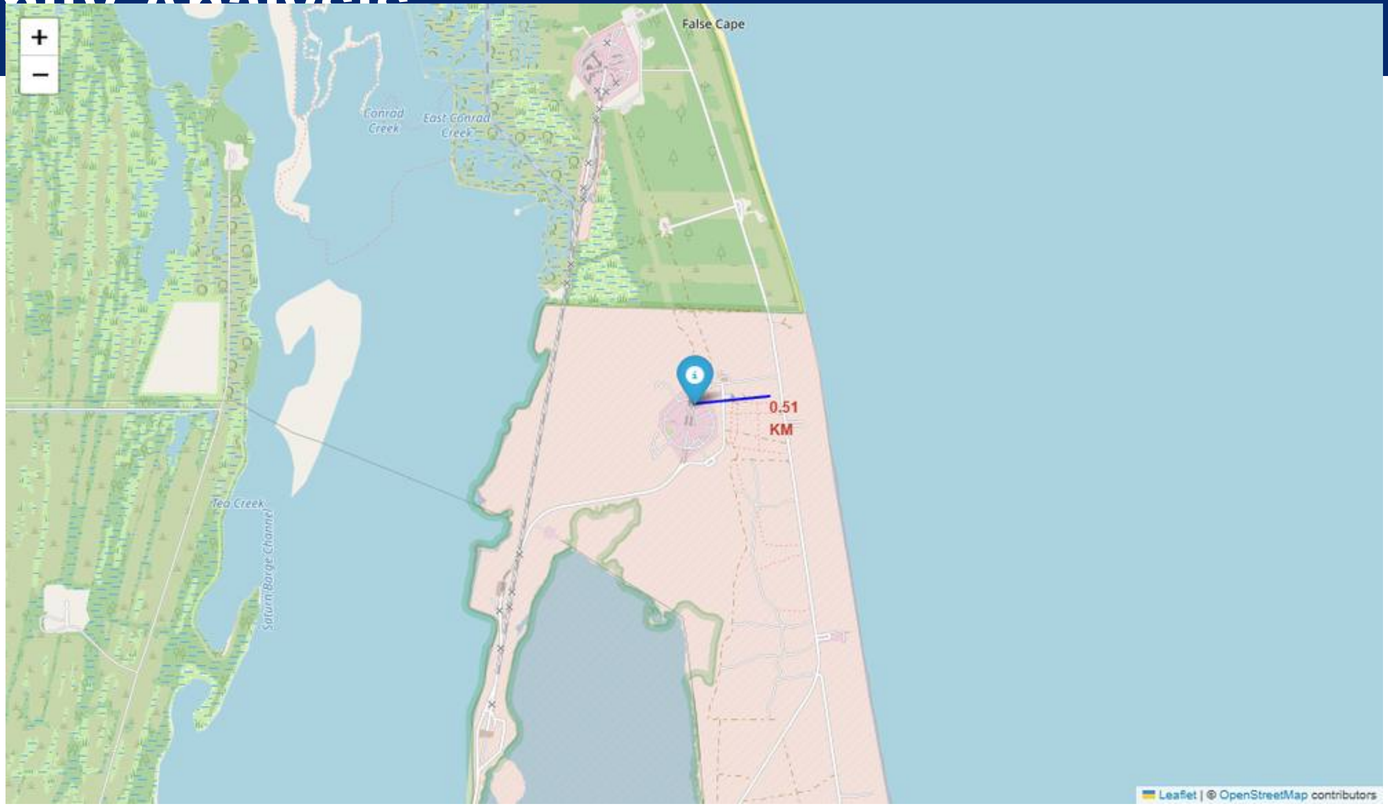
Insight: All sites are on US coasts - three in Florida (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and one in California (VAFB SLC-4E), close to the sea and near the equator.

Folium - Launch Outcomes by Site



Insight: Colour-coded markers (green = success, red = failure) clustered per site show which sites have the higher success ratio; KSC LC-39A stands out.

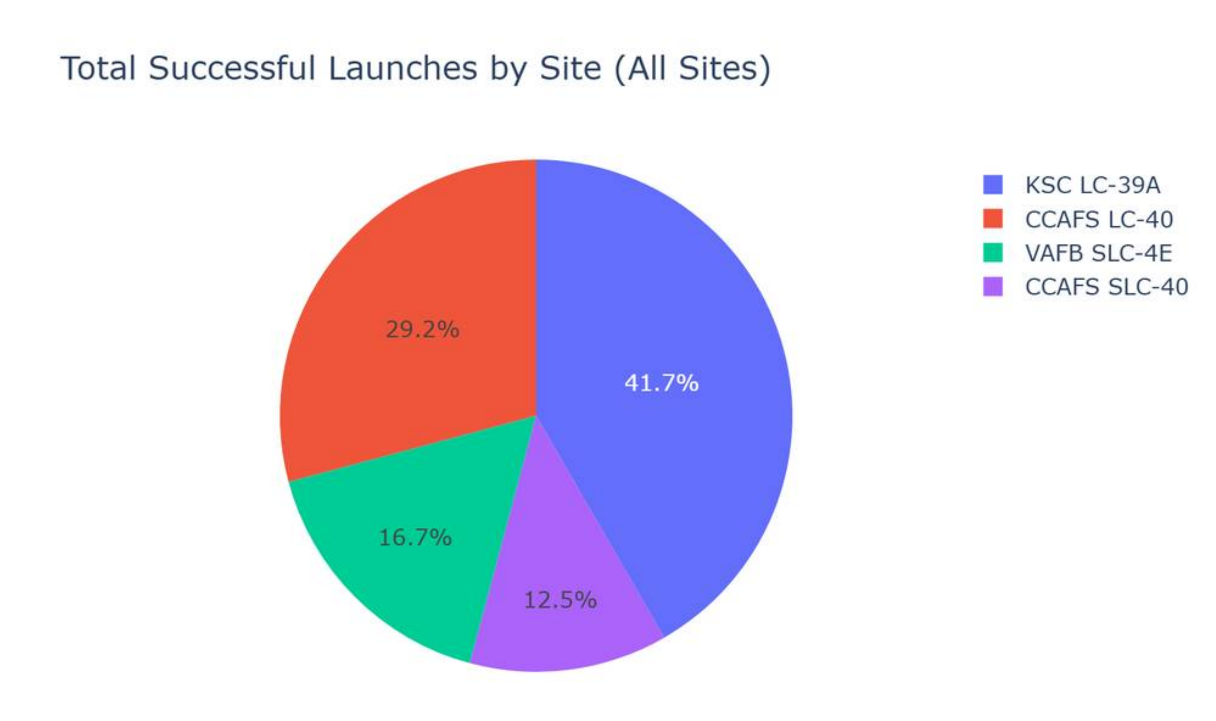
Folium - Proximity Analysis



Insight: Launch sites sit very close to the coastline (~0.51 km) and away from cities, consistent with launch-safety requirements.

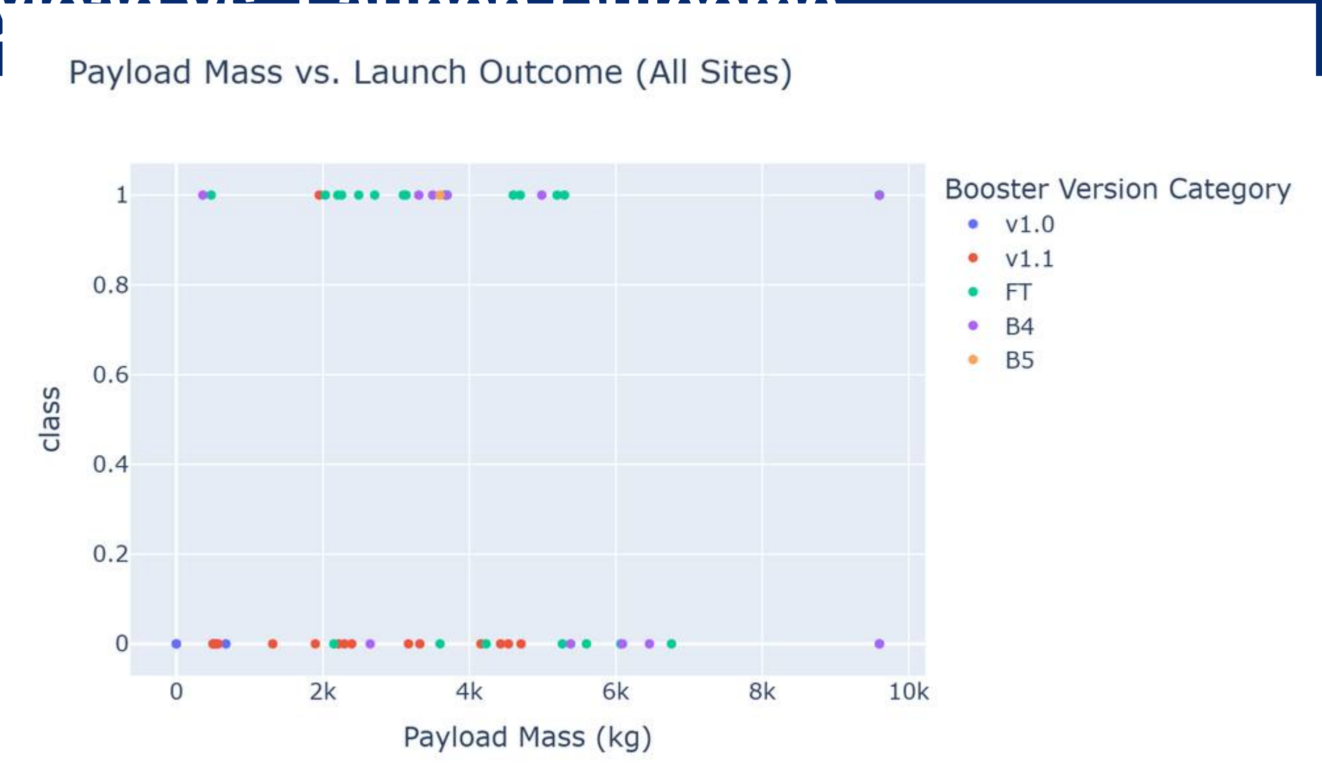
RESULTS - PLOTLY DASH

Dashboard - Successful Launches by Site



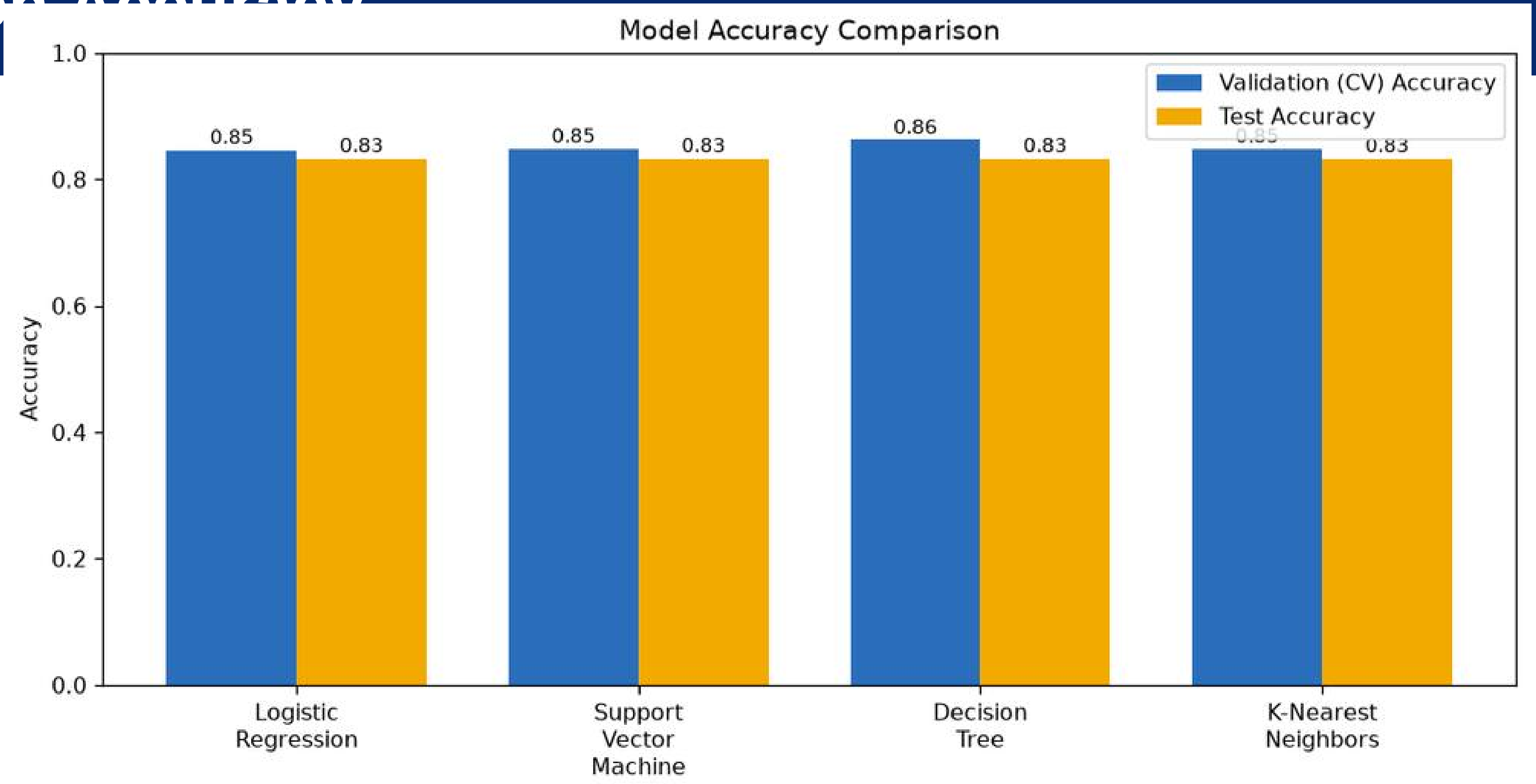
Insight: KSC LC-39A accounts for the largest share of successful launches, followed by CCAFS SLC-40.

Dashboard - Payload vs. Launch Outcome



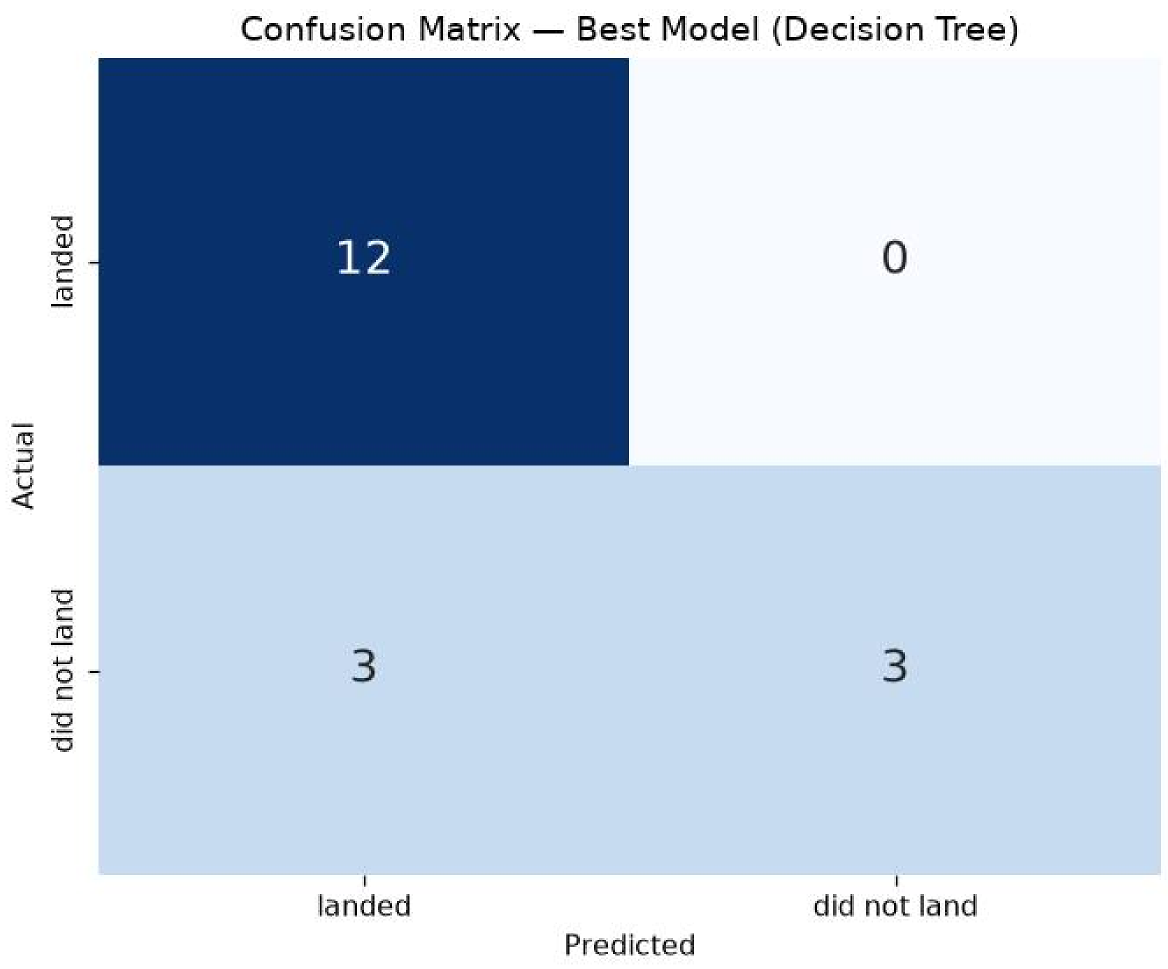
Insight: Low payload ranges show a higher success ratio than heavy payloads; the FT booster category has the strongest record.

Classification Accuracy



Insight: All four tuned models reach 83.3% test accuracy. The Decision Tree has the highest validation (CV) accuracy at 0.8625, making it the best model.

Confusion Matrix - Best Model (Decision Tree)



Reading the matrix

- True Positives (landed, predicted land): **12**
- True Negatives (did not land, predicted not): **3**
- False Positives: **3** ; False Negatives: **0**
- Accuracy = (12 + 3) / 18 = **83.3%**
- The main error is **false positives** - predicting a landing that did not occur.

Conclusions

- The Falcon 9 first-stage **landing success rate is ~67%** and has **improved dramatically since 2013**.
- **Launch site, orbit type and payload mass** are the strongest drivers of success; KSC LC-39A and low payloads land most reliably.
- Orbits **ES-L1, GEO, HEO, SSO** have the best landing outcomes.
- All tuned classifiers achieve **83.3% test accuracy**; the **Decision Tree** is the best model and is recommended for predicting landing cost.
- **Innovative insight:** because landing success (reusability) is now highly predictable, a competitor can reliably estimate SpaceX's true per-launch cost and price its bids accordingly.

Appendix

- Dataset: SpaceX REST API + Wikipedia web scraping (90 Falcon 9 launches).
- Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, SQLite/SQL, Folium, Plotly Dash, scikit-learn.
- Models tuned with GridSearchCV (cv=10); train/test split test_size=0.2, random_state=2.
- All notebooks and the Dash app: https://github.com/xwarcriminalx/IDM_Data_Coursera